

BURHANUDDIN SHIROSE

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Education

Carnegie Mellon University

Master of Science in Mechanical Engineering, Research: Motion Planning for Robotics

May 2024

CGPA - 3.90/4

National Institute of Technology, Tiruchirappalli

Bachelor of Technology in Mechanical Engineering

May 2022

CGPA - 8.41/10

Research Experience

Biorobotics Lab, Carnegie Mellon University

Advisor: Prof. Howie Choset

Scalable Multi-Agent Allocation for Multi-Objective Ergodic Search (MOES)

Mar 2026 – Present

- Reformulated MOES as clustering in Fourier space, exposing geometric structure for scalable task allocation
- Derived an infinite-horizon approximation to warm-start and accelerate finite-horizon task allocation.
- Developed a lazy ILP exploiting geometric structure for up to 97.2x speedup without sacrificing min-max optimality.
- Manuscript in preparation for submission to IEEE Robotics and Automation Letters.

Hierarchical Motion Planning for Snake Robot Reaching

Oct 2025 – Mar 2026

- Reduced high-dimensional snake robot planning to hierarchical search over compact geometric abstractions.
- Represented the robot with Bézier curves, reducing planning from 16 to 6 dimensions while preserving feasible motion.
- Validated the planner with a 98% simulation success rate and successful deployment on a physical 16-DoF snake robot.
- Research findings to be submitted in IEEE Robotics and Automation Letters.

Connectivity-Aware Coverage Motion Planning

Jan 2025 – Oct 2025

- Replaced frontier exploration with topology-guided planning over sparse connectivity graphs.
- Maintained a compact connectivity graph capturing observed and unobserved space to enable long horizon planning.
- Reduced total coverage time by 20%, shortened total traversal distance by 15%.
- Published research findings in the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Graph-Based Ergodic Search in Cluttered Environments

Aug 2023 – Mar 2024

- Replaced trajectory optimization with graph search over geometric free-space abstraction of the environment
- Designed an ergodicity-guided heuristic for informative, collision-free search over the generated graph.
- Validated the algorithm across 125 diverse scenarios with varying levels of clutter and different information distributions
- Published research findings in the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Engineering Experience

Robotics Engineer II – Carnegie Mellon University

Jan 2023 – June 2025

Motion Planning and Autonomous Navigation

Pittsburgh, PA

- Designed a real-time planning architecture for autonomous ground vehicles operating at 6 m/s.
- Accelerated online collision evaluation using efficient geometric footprint representations supporting 20 Hz replanning.
- Implemented advanced driver-assistance systems, including teleoperation assist and intelligent waypoint navigation.

Multi-Robot Planning

Pittsburgh, PA

- Developed decentralized planning architecture enabling scalable coordination across heterogeneous robotic teams.
- Represented formation objective through distributed coordination policies supporting scalable multi-robot assembly.
- Extended system to leverage heterogeneity, successfully coordinating mixed fleets of wheeled and quadruped robots.

Robotic Manipulation and Hardware Integration

Pittsburgh, PA

- Developed a synchronized multi-arm workcell comprising 4 UR10e manipulators for coordinated task execution.
- Developed a high-fidelity Hardware-in-the-Loop testbed to emulate microgravity docking for spacecraft physics research.
- Implemented velocity control to achieve sub-millimeter, sub-millisecond accuracy across multiple manipulator arms.

Technical Skills

Software Tools: Moveit, Drake, Pinocchio, Eigen, CasADi, Mujoco, Gazebo, Jax, ROS, OpenCV, Docker, Git

Publications

First Author

- **B. Shirose** et al. “GESCE: Graph-based Ergodic Search in Cluttered Environments” **IROS 2024**
- **B. Shirose** et al. “Robotic arm for brake performance testing” **RoAI 2021**

Middle Author

- Z. Shen, **B. Shirose** et al., “CAP: A Connectivity-Aware Hierarchical Coverage Path Planning Algorithm for Unknown Environments using Coverage Guidance Graph” **IROS 2025**
- C. Noren, S. Chaudhary, **B. Shirose** et al., “Communication Network Construction Behaviors for Robotic Convoying” **GVSETS 2025**
- P. Sriganesh, **B. Shirose** et al., “A Bayesian Modeling Framework for Estimation and Ground Segmentation of Cluttered Staircases” in **IEEE RA-L**
- C. Noren, B. Vundurthy, S. Scherer, M. Travers and **B. Shirose** “An interaction-aware two-level robotic planning and control system for vegetation override” **ISTVS 2024**
- B. Vundurthy, G. Gutow, A. Srinivasan, A. Xu, **B. Shirose** et al., “Generalized Multi-Agent Multi-Objective Ergodic Search” **AAAI MAPF Workshop**
- P. Sriganesh, J. Maier, A. Johnson, **B. Shirose** et al., “Modular, resilient, and scalable system design approaches- lessons learned in the years after DARPA subterranean challenge” **IEEE ICRA Workshop on Field Robotics, 2024**
- N. Kizor and **B. Shirose** et al. “Design of a Remotely Operated Vehicle (ROV) for Biofoul Cleaning and Inspection of Variety of Underwater Structures” **ICRoM 2021**